

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	02 October 2026
Team ID	SWTID-2026-3464
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Profile & Fitness Input	Enter name, age and weight Select fitness goal Select workout intensity Submit fitness information
FR-2	AI Workout Plan Generation	Generate personalized 7-day workout plan Generate goal-based exercises Adjust exercises according to workout intensity Use Gemini AI to generate the plan
FR-3	Nutrition & Recovery Guidance	Provide nutrition tips based on fitness goal Provide recovery guidance Provide rest-day recommendations
FR-4	Workout Plan Feedback	User can submit feedback on generated plan Modify workout plan based on feedback Regenerate updated plan using Gemini AI
FR-5	Workout Plan History	Store generated workout plans View previously generated plans Retrieve user workout information from database
FR-6	Admin Management	View registered users View generated workout plans Manage stored user information

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should provide a simple and user-friendly interface for entering fitness details and viewing workout plans.
NFR-2	Security	User information and stored workout plans should be protected from unauthorized access
NFR-3	Reliability	The system should generate workout plans consistently and handle invalid or incomplete inputs properly.
NFR-4	Performance	The application should respond efficiently when generating and displaying AI-based workout plans.
NFR-5	Availability	The FitBuddy application should be available to users whenever the application server is running.
NFR-6	Scalability	The system should be designed so that additional users and fitness features can be supported in the future